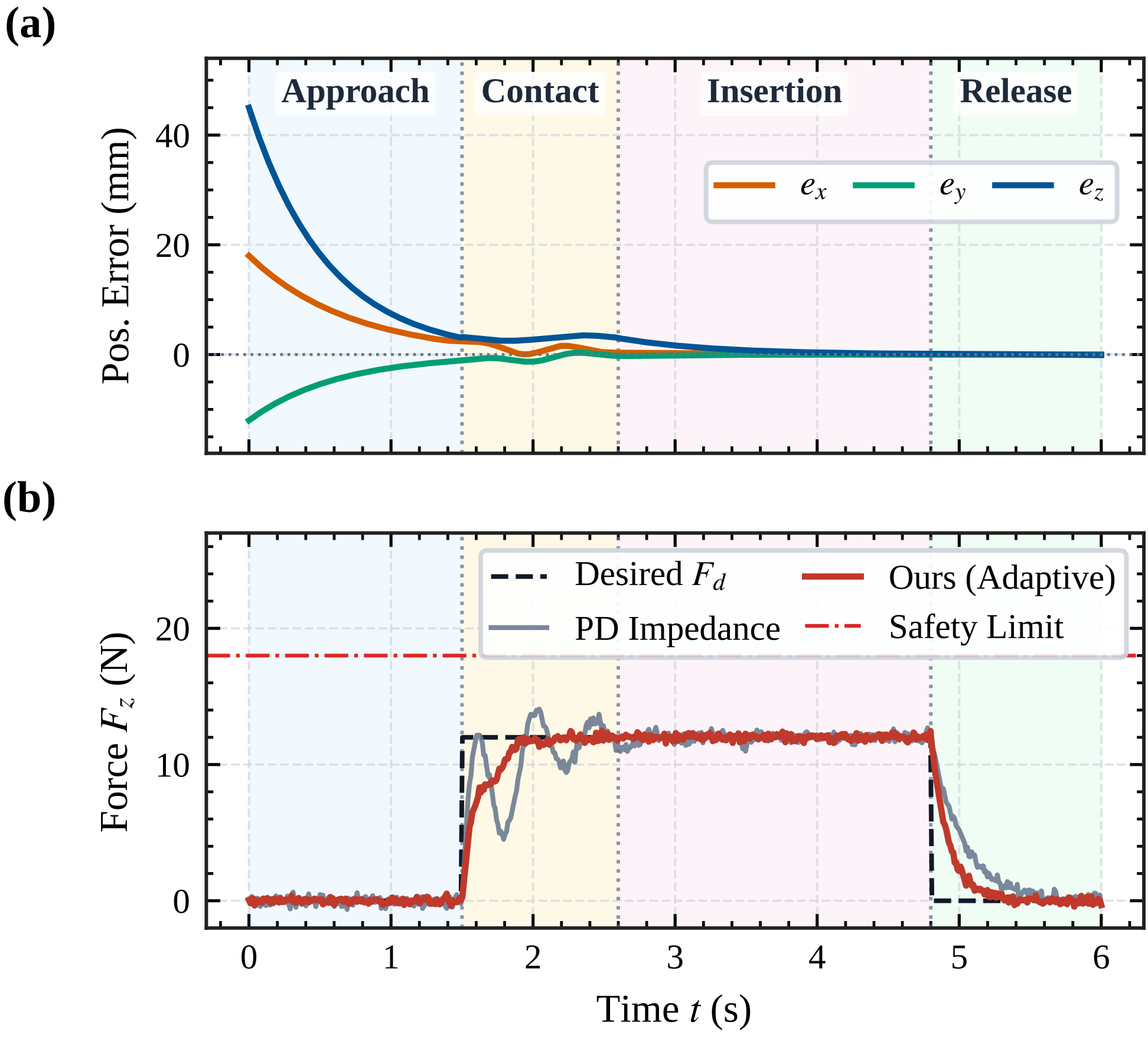
(a)
Approach
Contact
Insertion
Release
Pos. Error (mm)
40
20
0
e_x
e_y
e_z
(b)
Desired F_d
Ours (Adaptive)
PD Impedance
Safety Limit
Force F_z (N)
20
10
0
0
1
2
3
4
5
6
Time t (s)